

Trained prototype: what works now

Measured pollution forecasts + explicitly simulated plume and spraying

1 October 2026 | Original laboratory model unchanged | Physical field validation pending

MEASURED +30s

Outdoor PM2.5 / PM10 / temperature / RH

MEASURED +1/3/6h

Hourly pollution + weather inputs

SIMULATED CONTROL

PINN plume / DQN duties / toy efficiency

REAL-DATA MODELS

Short forecasts
New outdoor model uses 120 seconds of causal PM2.5, PM10, temperature and humidity history to forecast +30s.

Weather-aware forecasts
Separate hourly models forecast PM at +1/3/6h; wind, temperature and derived RH at +1h. Hourly weather cannot validate 30s predictions.

Risk and intervals
Learned 1/3/6h exceedance curves and residual-calibrated intervals. Thresholds are illustrative; coverage is reported.

Actual held-out checks
3,882 outdoor windows and 83,032 station-hours. Validation/calibration precede test; no old laboratory test used for fitting.

TRAINED SIMULATION

Plume PINN
Neural model trained with advection/diffusion/removal physics loss. Normalized idealized pulse; approximate footprint.

Spraying DQN
31 OFF/50%/100% zone actions. Fewer exposure seconds than reactive, but more water and worse combined reward. Reactive stays default.

Capture and arrival
Learned toy droplet/particle capture, simulated crossing-probability curves and arrival estimates. No measured treatment or exact ETA.

Portable exports
Plume/policy ONNX plus 12.2 KB int8 controller execute on desktop CPU. Serving uses full precision; board deployment is pending.

LIMITS / FUTURE WORK

Forecast weaknesses
Short PM2.5 loses baseline MAE. Its nominal 90% interval covered only 67.6%. Short RH loses persistence; short temperature selects persistence.

Fault detection
Learned anomaly flag caught 32.1% of injected extreme cases. It cannot reliably diagnose clogged or failing sensors.

Site validation
Need aligned dust/weather/position, nozzle duty, actual flow, treatment response and particle/droplet measurements; then physical trials.

Deployment / updates
Guarded fresh-data candidate retraining exists. Continuous field feed, reviewed model promotion and device performance tests remain.

MEASURED BENCHMARKS

MAE, ug/m3 (lower is better). Each comparison uses its own held-out data.

Task	Forecast	No change	Test windows
Outdoor PM10 +30s	6.310	7.300	3,882
Outdoor PM2.5 +30s	1.412	1.340	3,882
Hourly PM10 +1h	14.725	15.377	83,032
Hourly PM2.5 +1h	8.837	9.530	83,032

Different data from the original lab task: absolute errors do not prove improvement over that model.

SIMULATION TRADEOFF / 40 CASES, 10 MINUTES EACH

DQN: 4.896 L / 23.625 zone-seconds above setting. Reactive: 2.707 L / 70.500.
Continuous: 20.000 L / 20.500. Assumed flow and response; these are not field water savings.

TEAMMATE: run experimental/serve.py (Python 3.13, port 8011); use experimental/examples/.

Keep hourly/simulation labels. Unavailable models return 503; no fabricated live fallback.

Data: Askarov & Choi (2024), Mendeley; Chen (2017), UCI. Both CC BY 4.0.

Full evidence and setup: github.com/arifshekhk8/DustTwin-Model-Integration / experimental